

Equipotential Lines

ENGS124 Electricity and Magnetism Lab

21st October 2025

Introduction

The experiment is all about examining how electric potential and electric field are related and how electric forces act in space. Equipotential lines connect points of equal potential and make the electric field perpendicular to those lines. In this experiment we will examine three configurations of equipotential maps and observe how each has its unique field patterns. The goal of the experiment is to compare the results with theoretical prediction and confirm whether the electric field is equal to negative of the gradient of potential.

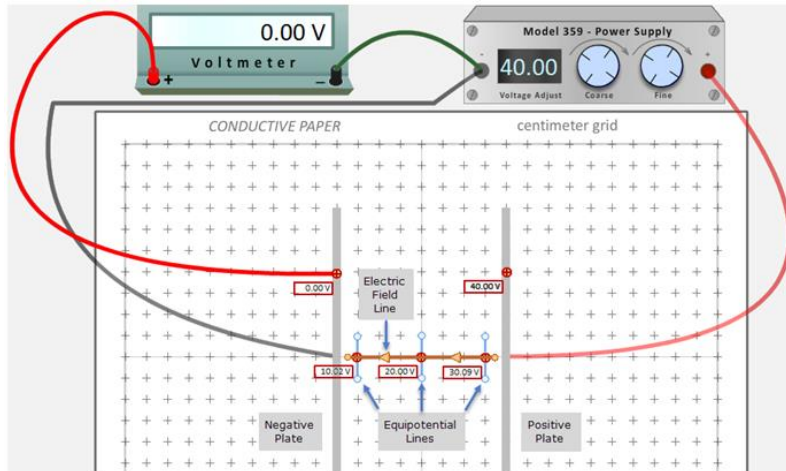
Total Errors

Name	Type	Impact	Solution
Factory Imperfection of Voltmeter	Systematic	Multimeter's voltage resolution and calibration may limit precision and shift all readings slightly.	Use a calibrated multimeter with finer resolution or verify calibration with known voltage reference.
Sudden Voltage Change Near metal surfaces	Systematic	The sudden voltage change near metal surfaces will give incorrect results during calculations, for example when calculating the gradient.	Record one extra measurement trying to decrease the neighborhood from the electrode or just exclude the point from calculations.
Geometry	Systematic	The geometry of the electrodes is assumed to be infinitely long or perfectly circular, but it is not, and the readings might slightly alter and change equipotential shapes or potential gradients.	Use larger electrodes. Move probes only perpendicular to the electrodes or just use perfect simulations.
Alignment Error	Systematic	If the measurement was not taken along the axis between electrodes, results slightly deviate.	Align the measurement path straight through the electrodes.

Part 1 (Linear Set Up)

This part uses two parallel electrodes in the water tray to produce a field that resembles the uniform field of a parallel plate capacitor.

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Plates	2	Good conducting metallic plates
Grid Paper	1	Grid paper with grid unit length of 1 cm
Connection Wires	4	Alligator wires
Probe	1	Metallic probe
Bowl With Water	1	Water with a depth of 1 cm

Theoretical Formula

$$E \equiv -\nabla V$$

$$E_x = -\frac{\Delta V}{\Delta d}$$

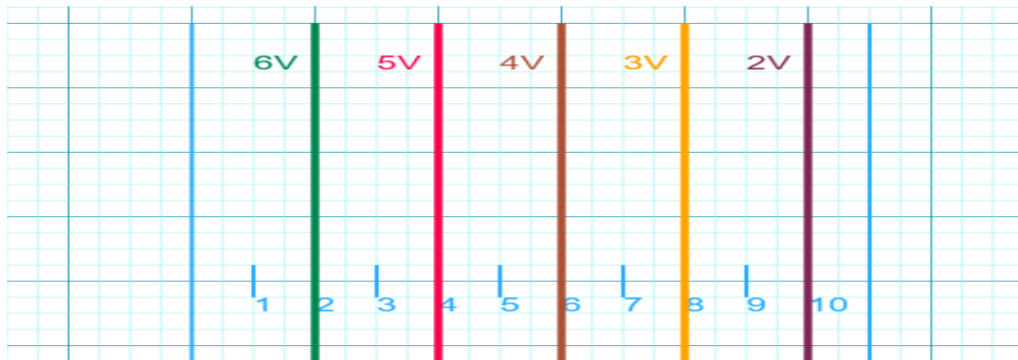
$$V(x) = \int_0^x \frac{\sigma}{\epsilon} dx = \frac{\sigma}{\epsilon} x = \frac{Q}{\epsilon A} x = \frac{V}{d} x$$

Data

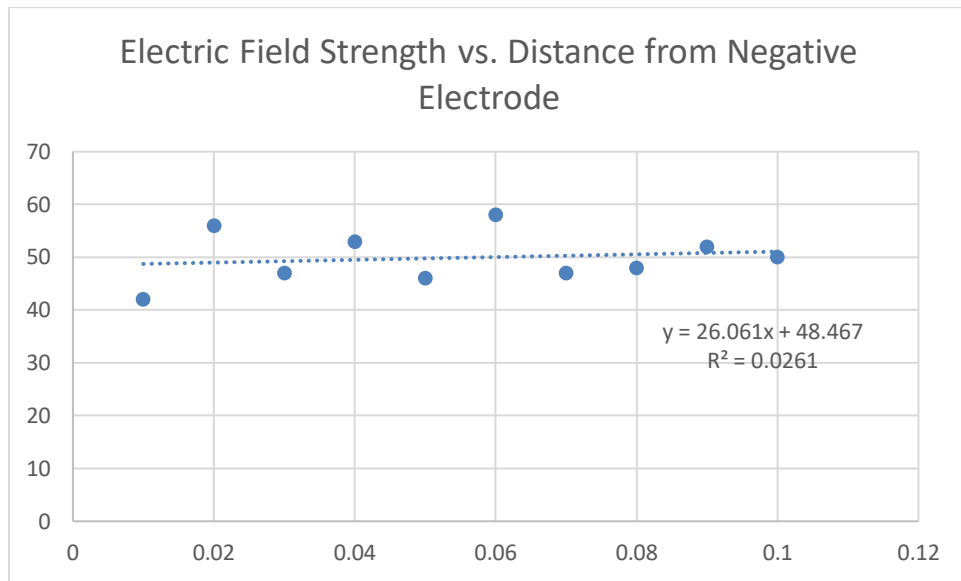
Distance from negative electrode, m (control)	V potential (measure)	$E_x = -\frac{\Delta V}{\Delta d}, V/m$ (calculate)
0	7	N/A
0.01	6.58	42
0.02	6.02	56
0.03	5.55	47
0.04	5.02	53
0.05	4.56	46
0.06	3.98	58
0.07	3.51	47
0.08	3.03	48
0.09	2.51	52
0.1	2.01	50

$$E_x = -\frac{\Delta V}{\Delta d}$$

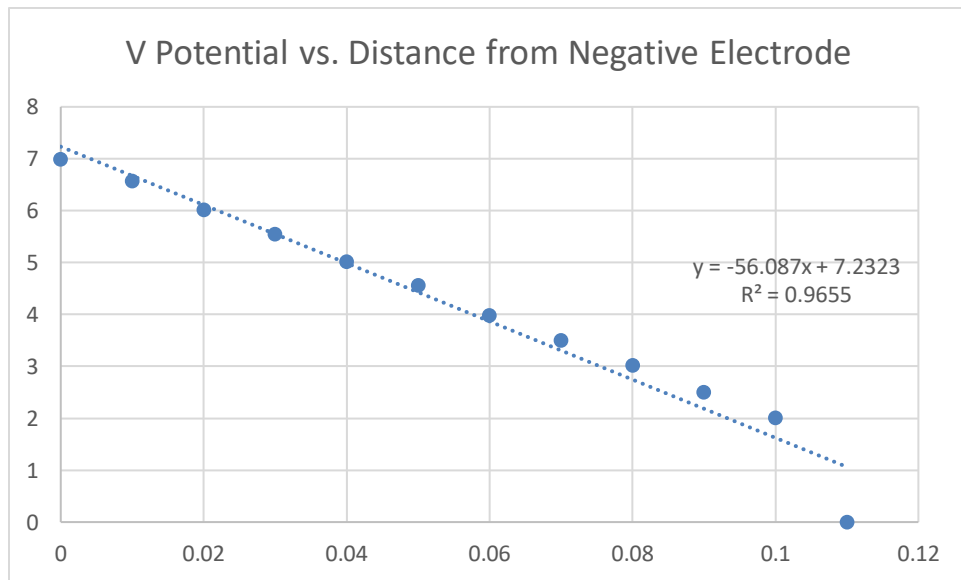
From this formula it is obvious that the closer the equipotential lines, the larger the electric field, and vice versa.



Plots



In upper graph there is no correlation between the data points, however the value of electric field strength is nearly constant, which implies that the field strength is uniform throughout the system.



In this graph in comparison to the previous one, there exists some strong correlation between data points, which implies that voltage changes linearly depending on distance from the electrode.

Variables

Name	Type	Resolution
d distance from negative electrode	Control	0.001 m
V potential	Measure	0.001 V
$E_x = -\frac{\Delta V}{\Delta d}$	Calculate	0.001 V/m

Comparing the electric field dependence curve with the theoretical one.

$$E_{theory} = \frac{\Delta V}{plate\ spacing} = \frac{7}{12} \approx 58.33V/m$$

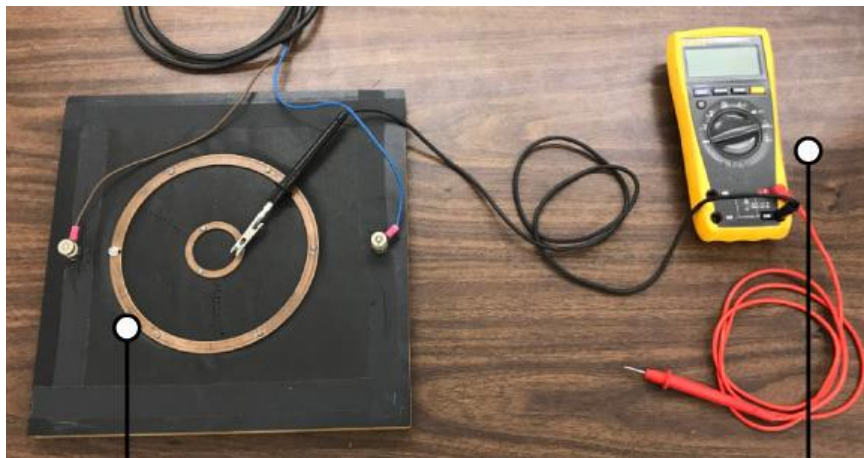
$$\mu(E_{experimental}) \approx 63.64\text{ V/m}$$

Percent error of 8.34%

Part 2 (Concentric Circle Set Up)

This part uses a central circular electrode and outer ring to create a radial electric field, which resembles a positively charged infinite straight rod.

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Outer conducting ring	1	Metallic ring, that is a good electricity conductor.
Grid Paper	1	Grid paper with grid unit length of 1 cm
Connection Wires	4	Alligator wires
Central Electrode	1	Metallic probe
Bowl With Water	1	Water with a depth of 1 cm

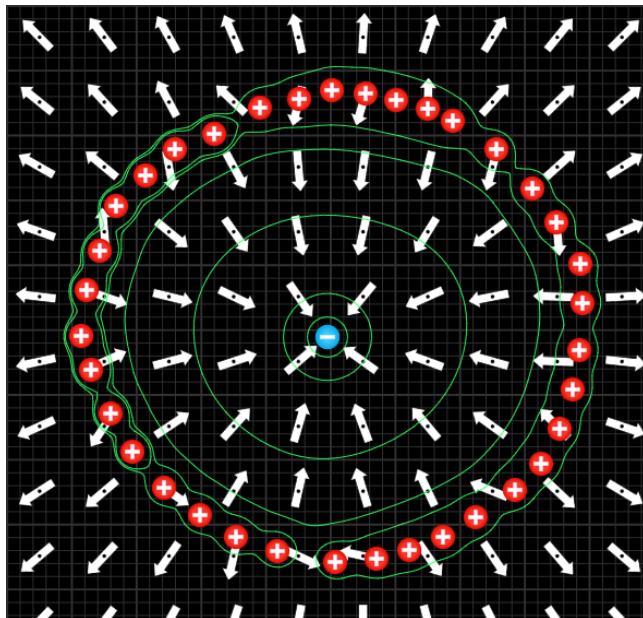
Theoretical Formula

$$E_r = \frac{\lambda}{2\pi\epsilon r}, \quad V(r) = \frac{\lambda}{2\pi\epsilon r} \ln \frac{r_0}{r} \quad (r \text{ can be neglected})$$

Linearization

$$E = -\frac{d}{dr}(\ln r) \sim \frac{1}{r}, \quad V \sim \ln r$$

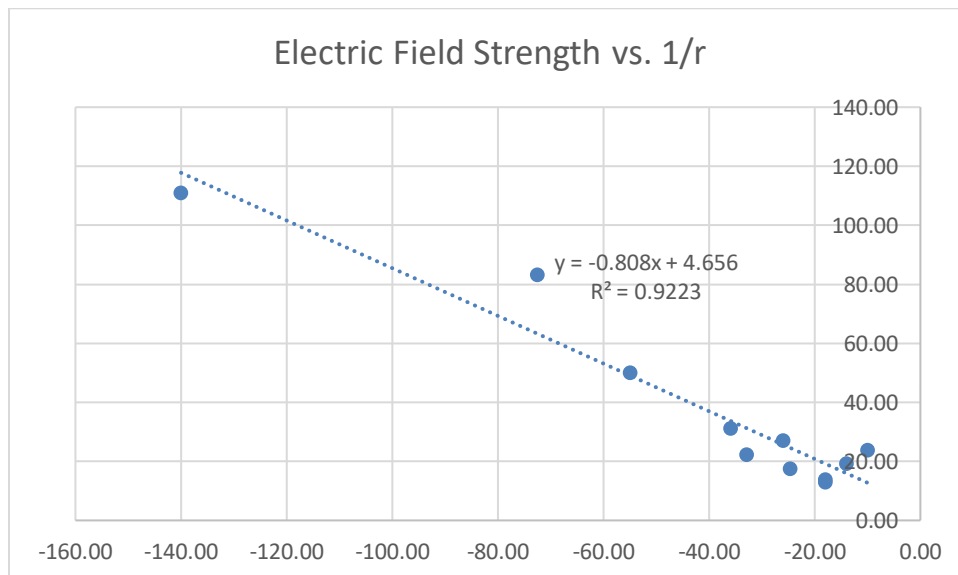
From the formula it is visible that the wider the spacing, the weaker the electric field.



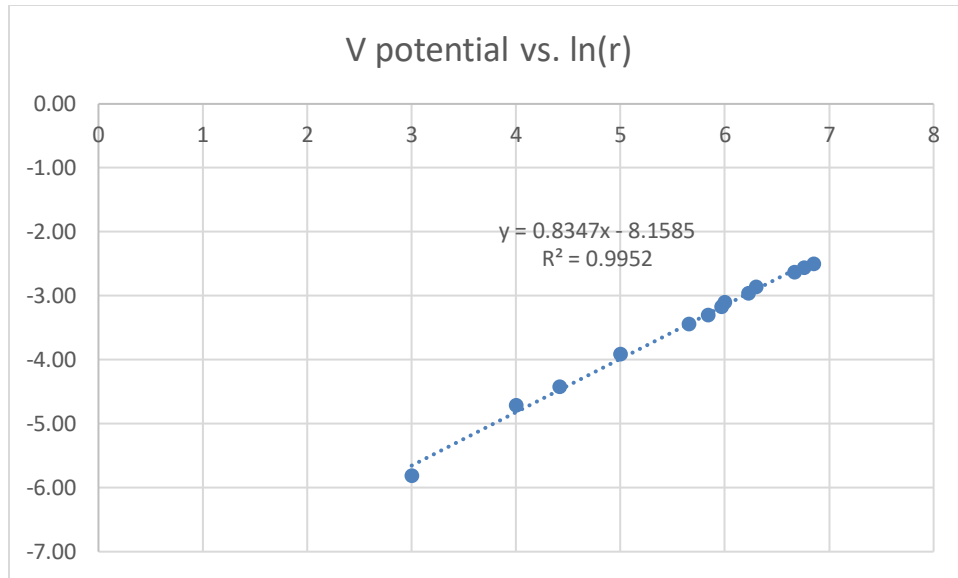
Data

d	V	$E_r = -\frac{\partial V}{\partial r}$	1/r	ln(r)
0.003	3	-166.67	333.33	-5.81
0.009	4	-140.00	111.11	-4.71
0.012	4.42	-72.50	83.33	-4.42
0.02	5	-55.00	50.00	-3.91
0.032	5.66	-36.00	31.25	-3.44
0.037	5.84	-26.00	27.03	-3.30
0.042	5.97	-10.00	23.81	-3.17
0.045	6	-32.86	22.22	-3.10
0.052	6.23	-14.00	19.23	-2.96
0.057	6.3	-24.67	17.54	-2.86
0.072	6.67	-18.00	13.89	-2.63
0.077	6.76	-18.00	12.99	-2.56
0.082	6.85	-83.54	12.20	-2.50

Plots



As we can see there is a strong linear relationship between the electric field strength and inverse of the radius, hence it implies that the electric field strength is inversely proportional to the radius. **The electric field strength decreases inversely with distance, stronger near rod.**



Once again, a strong linear relationship is visible, this time within the potential and natural logarithm of r , hence it implies that the V potential is proportional to the natural logarithm of radius. **V potential decreases slowly with distance.**

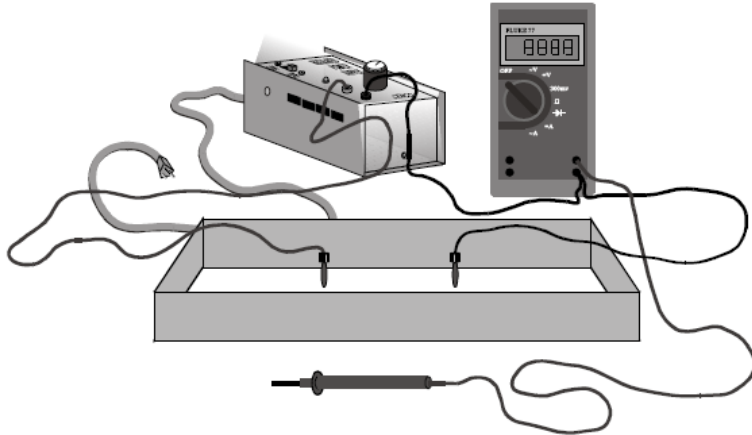
Variables

Name	Type	Resolution
r radius from negative electrode	Control	0.001 m
V potential	Measure	0.001 V
$E_r = -\frac{\Delta V}{\Delta r}$	Calculate	0.001 V/m
$1/r$	Calculate	0.001 m
$\ln(r)$	Calculate	0.001 m

Part 3 (Two Points Set Up)

In this section of the experiment two small electrodes of opposite charge create a dipole field. The goal of the experiment is to visualize curved equipotential lines and measure how potential changes along the line connecting the charges.

Setup

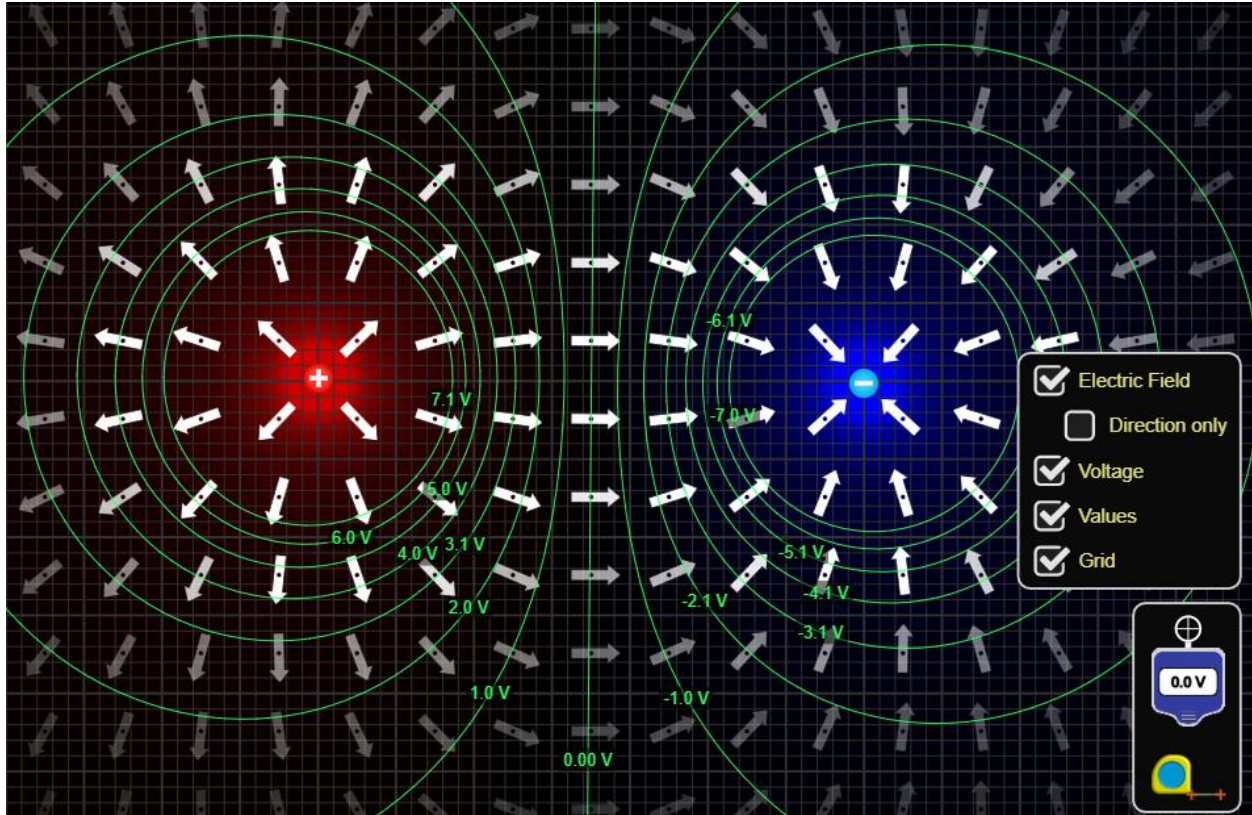


Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Probe	3	Good conducting metallic probes to stimulate point charges.
Grid Paper	1	Grid paper with grid unit length of 1 cm
Connection Wires	4	Alligator wires
Bowl With Water	1	Water with a depth of 1 cm

Theoretical Formula

$$E(r) = kQ \left(\frac{1}{r^2} + \frac{1}{(d-r)^2} \right), \quad V(r) = kQ \left(\frac{1}{d-r} - \frac{1}{r} \right)$$

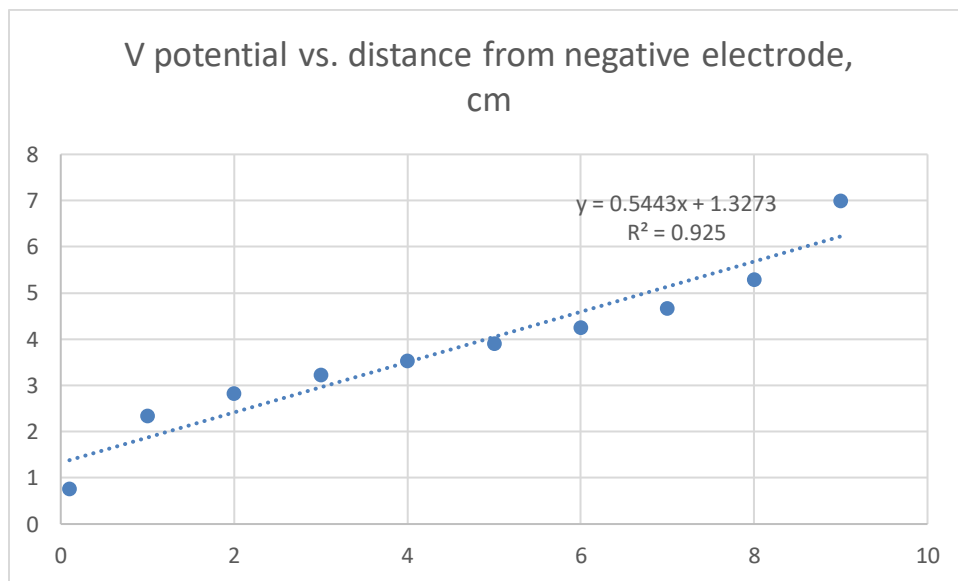
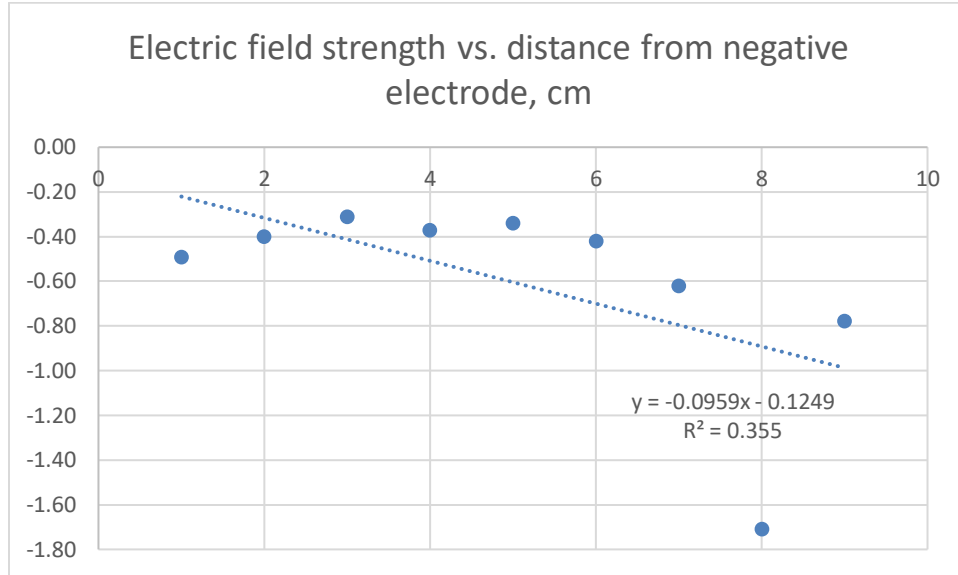


As we can see the equipotential lines appear to be more uniformly distributed on the line connecting two probes, however in other areas they start to behave differently.

Data

d	V	$E_r = -\frac{\partial v}{\partial r}$
0.1	0.76	N/A
1	2.34	-0.49
2	2.83	-0.40
3	3.23	-0.31
4	3.54	-0.37
5	3.91	-0.34
6	4.25	-0.42
7	4.67	-0.62
8	5.29	-1.71
9	7	-0.78

Plots



Strong relationship within V potential and distance from the negative electrode.

Potential increases as the probe is moved away from the negative electrode.

Variables

Name	Type	Resolution
r distance from negative electrode	Control	0.001 m
V potential	Measure	0.001 V
$E_r = -\frac{\Delta V}{\Delta d}$	Calculate	0.001 V/m